

Odometer DID Discovery on the INEOS Grenadier via Black-Box CAN Bus Reverse Engineering

Document type: Research findings

Vehicle: INEOS Grenadier MY23 LHD Diesel, VIN SC6GM1DA2PF005205 (vehicle #5205)

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Research platform: NVIDIA Jetson Orin Nano, SocketCAN, Kvaser Leaf v3, diesel plus ECUconnect V2.0B

App platform: OBDLink EX (macOS), OBDLink CX / vLinker MC+ (iOS BLE)

Protocol: ISO 15765-4, HS-CAN, 500 kbit/s, 11-bit addressing

Abstract

This document describes the discovery and validation of three independent odometer Data Identifiers (DIDs) on the INEOS Grenadier CAN diagnostic bus, accomplished through black-box reverse engineering of a Bosch Grade-X diagnostic session captured in parallel using a listen-only CAN bus sniffer. The methodology is entirely passive and non-invasive: no frames were injected, no ECU firmware was accessed, and no proprietary software was disassembled. The technique relies exclusively on observing the UDS (ISO 14229) communication between a legitimate OEM diagnostic tool and the vehicle's ECUs while the vehicle owner operates their own tool on their own vehicle. Three independent ECUs — BCM (Body Control Module), DDE (Digital Diesel Electronics), and HU (Head Unit) — were confirmed to independently store the vehicle odometer via UDS ReadDataByIdentifier (service 0x22) in the default diagnostic session, without requiring extended session access. All three ECUs returned the identical value of 99,939 km, cross-validated against the physical instrument cluster reading. The findings were implemented in an open-source iOS diagnostic application (`grenadier-obd2-ios`), enabling any Grenadier owner to read their own odometer via a standard Bluetooth OBD2 adapter.

1. Introduction

1.1 Motivation

The INEOS Grenadier is a purpose-built off-road vehicle produced since 2022, using BMW B57 (diesel) and B58 (petrol) powertrain technology. Its CAN diagnostic bus implements the standard BMW UDS protocol stack, making it architecturally similar to BMW 3-Series and X5 platforms of the same generation. Despite this, no public documentation of its specific diagnostic identifiers existed prior to this research project.

The vehicle odometer is one of the most fundamental data points a vehicle owner may wish to retrieve: for service history verification, insurance telematics, fleet management, or simply monitoring their vehicle's health via a custom application. Established OBD2 standards (SAE J1979 / ISO 15031) do not mandate an odometer PID in standard mode, and the Grenadier does not expose one via generic OBD2 broadcast. Odometer access on BMW-architecture vehicles therefore requires knowledge of BMW-proprietary UDS DIDs — which are not publicly documented.

This research demonstrates that such proprietary DIDs can be discovered legally, ethically, and reliably through systematic observation of legitimate diagnostic tool sessions using standard, commercially available hardware.

1.2 Research Question

Which ECUs on the INEOS Grenadier CAN diagnostic bus expose a vehicle odometer value via UDS ReadDataByIdentifier, and what are the precise DID addresses, data formats, and access conditions required to read them?

1.3 Significance

The discovery of three independent odometer DIDs — each accessible in default session without security access or extended session — is significant for the following reasons:

1. **Owner autonomy:** Vehicle owners can read their odometer via any standard ELM327 Bluetooth adapter without OEM tooling.
 2. **Fleet research:** Multiple independent ECU sources enable majority-vote validation, making the reading resilient to single-ECU faults or tampering.
 3. **Historical correlation:** DDE freeze-frame snapshots (UDS 0x19/0x04) contain the same odometer value as DID 0x46B1, confirming internal consistency across the ECU's own data model.
 4. **Open ecosystem:** The findings are implemented in an MIT-licensed iOS application, freely available to all Grenadier owners worldwide.
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2. Legal and Ethical Framework

2.1 The Right to Repair

The methodology described in this document is grounded in the well-established legal principle of the **vehicle owner's right to repair** — the right of a vehicle owner to access diagnostic information about their own vehicle without being compelled to use manufacturer-authorised tools or dealers.

This right is recognised and protected across all major jurisdictions in which the INEOS Grenadier is sold:

European Union — EU Regulation 2018/858 (Type Approval Framework)

Article 61 and Annex X mandate that vehicle manufacturers provide access to vehicle repair and maintenance information (RMI) to independent operators on a non-discriminatory basis. This includes OBD diagnostic data. A vehicle owner is the ultimate beneficiary of this regulation — if independent workshops have the legal right to diagnostic access, the vehicle owner has an even more fundamental right to monitor their own property.

European Union — EU Regulation 715/2007 (Euro 5/6 Emissions)

Article 6 requires manufacturers to provide standardised OBD access. Standard OBD2 protocols (ISO 15765) used in this research are the same protocols mandated by EU emissions regulation.

United Kingdom — The Road Vehicles (Approval) Regulations 2020

Post-Brexit UK retained equivalent provisions. UK consumers retain the right to access diagnostic information for their own vehicles and to have repairs performed by independent workshops using standard diagnostic protocols.

Australia — ACCC Right to Repair Framework

The Australian Competition and Consumer Commission explicitly supports the right of consumers and independent repairers to access vehicle diagnostic information. The ACCC's 2021 inquiry into Automotive data access concluded that vehicle owners have legitimate interests in accessing their vehicle's data.

United States — Massachusetts Right to Repair Act (2012) and 2020 Update

The original Massachusetts Right to Repair Act (and subsequent industry-wide memorandum of understanding) established that vehicle manufacturers must provide OBD data access via standard protocols

to vehicle owners and independent repairers. The 2020 update extended these provisions to telematics data.

2.2 Passive Observation is Not Circumvention

A critical legal distinction applies to this research: **observation of CAN bus traffic is fundamentally different from circumventing access controls.**

The methodology employed here is analogous to observing the radio signals broadcast by your own radio transmitter, or reading the network packets sent across your own LAN. No cryptographic protection was bypassed. No firmware was extracted, decompiled, or reverse engineered. No authentication mechanism was defeated.

The Bosch Grade-X diagnostic session was initiated legitimately by the vehicle owner using commercially available tooling. The CAN bus traffic generated by this session was observed on the vehicle owner’s own vehicle’s own CAN bus using hardware physically plugged into the vehicle owner’s own OBD2 port. This is not interception — it is observation of one’s own property.

Computer Fraud and Abuse Act (US) — 18 U.S.C. § 1030: Does not apply. There was no “unauthorised access to a computer” — the vehicle owner has full authorisation to observe diagnostic traffic on their own vehicle. The ECUs in question are not “protected computers” in the legal sense when accessed via the vehicle’s own diagnostic port by its owner.

EU Directive 2001/29/EC (Copyright): Does not apply. No software was copied, decompiled, or redistributed. The UDS protocol is an international standard (ISO 14229). The specific DID addresses and their data formats observed on the CAN bus are not copyrightable — they are facts about the vehicle’s behaviour, discovered by observation.

EU Directive 2016/943 (Trade Secrets): Does not apply. The information was not obtained by improper means — it was observed on the vehicle owner’s own CAN bus during the legitimate operation of their own vehicle. The EU Trade Secrets Directive explicitly does not protect information that can be obtained through legitimate observation or study.

2.3 Ethical Conduct

This research adheres to the following ethical principles:

- **No vehicle harm:** All observations were passive. No frames were injected during Grade-X sessions. No ECU was modified, flashed, or placed in an unexpected state.
- **No proprietary software redistribution:** The Bosch Grade-X software itself was not copied, distributed, or decompiled. Only the observable behaviour of the CAN bus was recorded.
- **Public benefit:** All findings are published openly so that any Grenadier owner may benefit from them. No finding is sold, licensed, or restricted.
- **Safety:** No findings enable safety-relevant modifications. Reading an odometer value carries zero safety risk.

3. Technical Background

3.1 CAN Bus Architecture on the INEOS Grenadier

The INEOS Grenadier exposes a single diagnostic CAN bus at the OBD2 port (SAE J1962 connector, pins 6 and 14). This is a **high-speed CAN (HS-CAN) bus operating at 500 kbit/s with 11-bit addressing**, implementing ISO 15765-4 transport.

This diagnostic bus is not the primary powertrain CAN (PT-CAN) — it is a dedicated diagnostic bus. Standby traffic consists of three always-present IDs:

CAN ID	Rate	Purpose
0x7F0	~5 ms	Network Management heartbeat (two interleaved sources)
0x03C	100 ms	Vehicle master clock / time base
0x130	100 ms	Gateway NM acknowledgement

All ECU communication uses functional addressing via UDS (ISO 14229), with each ECU assigned a fixed request/response address pair (physical addressing). The addressing scheme follows BMW conventions: request addresses in the range 0x680–0x6CF, response addresses are request + 0x080 (i.e., 0x600–0x64F).

3.2 UDS Protocol — Service 0x22 ReadDataByIdentifier

UDS service 0x22 (ReadDataByIdentifier) is the primary mechanism for reading fixed data from an ECU. A request specifies a 2-byte Data Identifier (DID); the ECU responds with the DID's current value.

Request frame structure (CAN, ATCAF0 mode, D1-padded):

CAN ID	PCI	SID	DID_H	DID_L	[padding]
681	03	22	F1	21	D1 D1 D1 D1

Positive response:

CAN ID	PCI	SID+40	DID_H	DID_L	[data bytes]
601	06	62	F1	21	01 86 63 00

The positive response service byte is always SID + 0x40 — in this case 0x22 + 0x40 = 0x62.

3.3 Default Session vs. Extended Session

UDS defines multiple diagnostic sessions: - **Default session (0x01):** Always active. Limited DID access. - **Extended diagnostic session (0x03):** Unlocks additional DIDs and services. Requires 10 03 DSC request. - **Programming session (0x02):** Requires security access (0x27 seed/key exchange).

A critical finding of this research is that **all three odometer DIDs are accessible in the default session** — no session escalation, security access, or keepalive is required. The ECU responds immediately to a cold 22 DID request.

3.4 ISO-TP Framing (ISO 15765-2)

For responses longer than 7 bytes, ISO 15765-2 multi-frame transport is used:

Frame Type	PCI Nibble	Purpose
Single Frame (SF)	0x0N	Complete payload in one frame (N = length)
First Frame (FF)	0x1N NN	Start of multi-frame; total length encoded in 12 bits
Consecutive Frame (CF)	0x2N	Continuation; N = sequence number 1–F

Frame Type	PCI Nibble	Purpose
Flow Control (FC)	0x3N	Receiver grants/throttles transmission

All three odometer DIDs return single-frame responses (payload ≤ 7 bytes), meaning no flow control is needed and the round-trip is a single request/response pair.

4. Research Methodology

4.1 Overview: Black-Box Testing via Parallel CAN Capture

The core methodology is **black-box observation**: a legitimate OEM diagnostic tool (Bosch Grade-X) operates normally on the vehicle while a passive, listen-only CAN sniffer records all traffic simultaneously. The result is a complete transcript of every UDS request and response exchanged between the diagnostic tool and every ECU in the vehicle.

This approach is powerful because:

1. **Ground truth:** Grade-X is the OEM tool — whatever it reads is definitionally correct. If Grade-X reads a DID and the value matches the physical instrument cluster, the DID is confirmed.
2. **Completeness:** Grade-X queries many DIDs not previously investigated. The capture reveals these queries and their responses automatically.
3. **No guessing:** There is no need to iterate through the 65,536 possible DID addresses by trial and error — Grade-X queries the relevant ones directly.
4. **Validation:** Grade-X’s own queries provide the confirmation that a DID is meaningful to the OEM — not just an accident of ECU firmware.

4.2 Hardware Configuration

Two CAN interfaces were active simultaneously throughout the session:

Interface	Device	Role	CAN Mode
can0	diesel plus ECUconnect V2.0B (gs_usb)	Passive sniffer	listen-only — never transmits
can1	Kvaser Leaf v3 (kvaser_usb)	Active (prior/subsequent work only)	500 kbps TX/RX

Both adapters were connected to the vehicle OBD2 port simultaneously. The ECUconnect adapter was configured in **listen-only** mode at OS level:

```
sudo ip link set can0 up type can bitrate 500000 listen-only on
```

In listen-only mode, the adapter **does not generate ACK bits** on the CAN bus. It is electrically invisible to all other nodes — a pure passive observer. It cannot cause bus errors, cannot influence timing, and cannot affect ECU behaviour in any way.

The Bosch Grade-X PC software was connected via its own independent hardware interface (not captured here — Grade-X uses a proprietary adapter). Grade-X operated completely normally, unaware of the parallel capture.

4.3 Capture Process

CAN traffic was captured using `candump` with absolute timestamps:

```
candump -t a can0 | tee dumps/grade_x_sessions/grenadier_raw_20260619_154716.log
```

The `tee` command duplicates output to both the terminal (live monitoring) and a timestamped log file. The capture was started before Grade-X initiated communication and stopped after Grade-X disconnected. The resulting log file contains every CAN frame visible on the diagnostic bus for the duration of the session — both Grade-X’s requests and every ECU’s responses.

Session statistics for the primary session (grenadier_raw_20260619_154716.log): - Duration: **841 seconds** (~14 minutes) - Total frames captured: **243,075** - Active CAN IDs: all 19 known ECU request/response pairs, plus standby traffic - Notable operations captured: TPMS reprogramming, security access, DID writes, routine control, DTC reads, **odometer reads**

4.4 Analysis Process

Post-session analysis was performed on the raw log file using standard Unix command-line tools (`grep`, `awk`, `python3`) and the project’s existing analysis infrastructure.

The search for odometer-related traffic was triggered by the commit message of the Grade-X session commit, which explicitly mentioned “reading Odometer” as one of the operations performed. This directed analysis toward responses containing the known odometer value.

Step 1 — Determine the known odometer value

The vehicle owner confirmed the physical instrument cluster reading at the time of the session: **99,939 km**. This was independently verified as consistent with prior sessions: - DTC snapshot analysis (2026-06-17): EGS CF2E01 onset at 99,922 km - Grade-X session (2026-06-19): 99,939 km = 99,922 + 17 km driven between sessions OK

Step 2 — Convert to hexadecimal

```
km = 99939
print(hex(km))    # → 0x18663
# As 3-byte big-endian: 01 86 63
# As 4-byte big-endian: 00 01 86 63
```

Step 3 — Search all three Grade-X logs

```
grep "01 86 63" dumps/grade_x_sessions/grenadier_raw_20260619_*.log
```

This immediately revealed three distinct ECU response frames containing the odometer value, across the three capture files.

5. Technical Findings

5.1 Raw Frame Evidence

The search yielded hits in three separate ECU response streams, each from a different Grade-X log file:

HU response — grenadier_raw_20260619_152801.log:

(1781875082.260165) can0 603 [8] 06 62 06 0A 01 86 63 AA

BCM and DDE responses — grenadier_raw_20260619_153651.log (repeated 8× across session):

(1781875694.259474) can0 601 [8] 06 62 F1 21 01 86 63 00

(1781875694.260165) can0 612 [8] 07 62 46 B1 00 01 86 63

The repetition of BCM and DDE responses (8 occurrences each) corresponds to Grade-X re-reading these DIDs at multiple points in the session — after DTC clear, after security access, after write operations. Each re-read returned the identical value, confirming stability.

5.2 Frame Decoding

5.2.1 BCM — DID 0xF121

Frame: 06 62 F1 21 01 86 63 00
PCI SID DID DID D0 D1 D2 pad
(6) +40 hi lo

- PCI 06 = 6 data bytes follow
- SID 62 = positive response to service 0x22 (ReadDataByIdentifier)
- DID F1 21 = 0xF121
- Data: 01 86 63 = 3 bytes
- Trailing 00 = CAN frame padding (not part of the DID value)

Decode: 0x018663 = **99,939 km** (unsigned 24-bit big-endian integer)

5.2.2 DDE — DID 0x46B1

Frame: 07 62 46 B1 00 01 86 63
PCI SID DID DID D0 D1 D2 D3
(7) +40 hi lo

- PCI 07 = 7 data bytes follow
- SID 62 = positive response to service 0x22
- DID 46 B1 = 0x46B1
- Data: 00 01 86 63 = 4 bytes

Decode: 0x00018663 = **99,939 km** (unsigned 32-bit big-endian integer)

5.2.3 HU — DID 0x060A

Frame: 06 62 06 0A 01 86 63 AA
PCI SID DID DID D0 D1 D2 pad
(6) +40 hi lo

- PCI 06 = 6 data bytes follow
- SID 62 = positive response to service 0x22
- DID 06 0A = 0x060A
- Data: 01 86 63 = 3 bytes
- Trailing AA = CAN frame padding (not part of the DID value)

Decode: 0x018663 = **99,939 km** (unsigned 24-bit big-endian integer)

5.3 Consolidated DID Table

ECU	Full Name	TX (re- quest)	RX (re- sponse)	DID	Session	Data	Format	Value
BCM	Body Control Module	0x681	0x601	0xF121	Default	3 bytes	uint24 BE	99,939 km
DDE	Digital Diesel Electronics	0x692	0x612	0x46B1	Default	4 bytes	uint32 BE	99,939 km
HU	Head Unit	0x683	0x603	0x060A	Default	3 bytes	uint24 BE	99,939 km

5.4 DID Namespace Analysis

The three DIDs inhabit different namespaces within the BMW UDS DID allocation:

- **0xF121** — in the 0xF1xx range, which is the BMW-defined Vehicle Identification Data range (standardised sub-range). 0xF190 is VIN, 0xF121 is odometer. This range is consistently available in default session across BMW-architecture ECUs.
- **0x46B1** — in the 0x4xxx range, which is BMW-defined powertrain / engine management data. DDE/DME-specific. The 4-byte uint32 format accommodates future odometer values beyond 16,777,215 km (the 3-byte uint24 ceiling).
- **0x060A** — in the 0x0xxx range, which is BMW-defined infotainment / display data. The Head Unit maintains an odometer copy presumably for its service reminder and maintenance display functions.

The fact that all three DIDs exist across separate namespaces maintained by separate software teams (BCM firmware, DDE/DME powertrain firmware, HU infotainment firmware) is itself significant: it confirms that the odometer is a first-class shared datum in the BMW electrical architecture, synchronised across ECU boundaries via the internal vehicle CAN network.

5.5 Cross-Validation with DTC Freeze Frame Data

An independent confirmation of the odometer encoding was found within DTC freeze frame data (UDS 0x19/0x04) in the same capture session. DDE's freeze frame response for DTC 0xCD969F contained DID 0x2117 with value 00 01 86 63 = 99,939 km:

```
(1781877594.259474)  can0  612   [8]  10 98 59 04 CD 96 9F 27    ← FF: 152-byte snapshot
(1781877594.259903)  can0  692   [8]  30 00 00 FF FF FF FF FF    ← FC: continue all
(1781877594.260165)  can0  612   [8]  21 01 21 17 00 01 86 63    ← CF1: rec=01, DID=0x1217
```

DID 0x1217 in DDE freeze frames encodes the odometer at the time of fault occurrence — the same 4-byte uint32 BE format as 0x46B1. This internal consistency across DDE's own data model (live DID vs. freeze frame DID) provides strong corroborating evidence that the encoding is correct and the value is authoritative.

5.6 Access Conditions

A key finding is the absence of any access barrier to these DIDs:

Condition	Required?	Evidence
Extended session (0x03)	No	Grade-X reads these in default session; direct test confirms
Security access (0x27)	No	No seed/key exchange precedes the odometer reads
TesterPresent keepalive	No	No keepalive frames in window before odometer reads
Ignition ON	Yes	All three ECUs require ignition ON to respond; NRC 0x22 or NO DATA when off
Engine running	No	Ignition accessory position is sufficient

The requirement for ignition ON is consistent with all other diagnostic reads on this platform and does not represent an access barrier in any meaningful sense — any vehicle owner operating their own OBD2 reader will have their ignition in accessory position.

6. Majority Vote Architecture

6.1 Rationale

Reading the odometer from three independent ECUs and taking a majority vote provides resilience that a single-source read cannot:

- **ECU fault resilience:** If any single ECU fails to respond (power fault, bus error, firmware issue), two remaining sources still produce a valid majority.
- **Tampering detection:** Odometer fraud typically involves rolling back one ECU's value. Three independent sources with different software stacks and different DID namespaces make a consistent rollback across all three vastly more difficult. A mismatch between sources immediately flags a discrepancy.
- **Validation confidence:** When all three sources agree (as confirmed on vehicle #5205), confidence in the reading is extremely high.

6.2 Implementation

The majority vote algorithm in Swift:

```
func readOdometer() async {
    let bcmKm = await atLayer.readOdometerDirect(tx: "681", rx: "601", did: "F121")
    let ddeKm = await atLayer.readOdometerDirect(tx: "692", rx: "612", did: "46B1")
    let huKm  = await atLayer.readOdometerDirect(tx: "683", rx: "603", did: "060A")

    let readings = [bcmKm, ddeKm, huKm].compactMap { $0 }
    guard !readings.isEmpty else { return }
    let majority = readings.max(by: { a, b in
        readings.filter { $0 == a }.count < readings.filter { $0 == b }.count
    })
}
```

```

    }) ?? readings[0]

    self.vehicleOdometer = majority
}

```

The DID reader itself uses a generic decoder that handles both 3-byte and 4-byte odometer encodings:

```

func readOdometerDirect(tx: String, rx: String, did: String) async -> Int? {
    guard let payload = try? await sendUDS("22\$(did)", tx: tx, rx: rx),
          payload.count >= 5 else { return nil }
    // payload: [0x62, DID_H, DID_L, data...]
    let data = Array(payload.dropFirst(3))
    var value = 0
    for byte in data.prefix(4) { value = (value << 8) | Int(byte) }
    return value > 0 ? value : nil
}

```

The `prefix(4)` call handles both uint24 (3-byte BCM/HU payload → 3 iterations) and uint32 (4-byte DDE payload → 4 iterations) without branching.

7. Validation Results

7.1 Physical Instrument Cluster Confirmation

The vehicle owner confirmed the instrument cluster reading at the time of the Grade-X session:

“my Odometer shows exactly 99939 km for this Grade-X session, no more no less”

This provides direct, ground-truth validation of the decoded values. All three ECUs returned 99,939 km and the physical dashboard showed 99,939 km. Agreement is exact, not approximate.

7.2 Longitudinal Consistency

Cross-session odometer tracking further validates the findings:

Session	Date	Odometer	Source
DTC snapshot analysis (EGS CF2E01)	2026-06-17	99,922 km	ZF freeze frame DID 0x1517
DTC snapshot analysis (GWS latest)	2026-06-17	99,922 km	ZF freeze frame DID 0x0817
Grade-X session + direct DID read	2026-06-19	99,939 km	BCM/DDE/HU + dashboard
Delta	2 days	+17 km	OK consistent with normal use

The +17 km delta between the 2026-06-17 and 2026-06-19 sessions is consistent with normal vehicle usage (approximately one round trip of local driving). This longitudinal consistency confirms that the DID values are live odometer readings, not static or cached values.

7.3 Internal ECU Consistency

The DDE ECU independently stores the odometer in two separate locations within its firmware:

Location	DID	Format	Value
Live data	0x46B1	uint32 BE	99,939 km
Freeze frame DID 0x2117 (in 0x19/0x04 snapshot for DTC CD969F)	—	uint32 BE	99,939 km

Identical encoding, identical value. This demonstrates that the DDE’s odometer datum is consistent across its own internal subsystems — further evidence that 0x46B1 is the canonical live odometer DID for this ECU.

8. Practical Characteristics for Application Development

8.1 Recommended Read Sequence

For application developers targeting the INEOS Grenadier (or any BMW-architecture vehicle using equivalent ECUs):

1. **Confirm ignition ON** (BCM responds to 22F190 with FF+CFs when ignition ON, NRC 7F 22 22 when OFF)
2. **Read BCM 0xF121** — fastest, single-frame, always available first
3. **Read DDE 0x46B1** — 4-byte uint32, handles values > 16.7M km
4. **Read HU 0x060A** — third independent source for majority vote
5. **Apply majority vote** across three readings
6. **Log all three raw values** for fleet analytics and anomaly detection

8.2 ELM327 AT Command Sequence

For ELM327-based adapters (OBDLink, vLinker, etc.) in ATCAF0 mode:

```
ATSH 681      → set TX header to BCM
ATCRA 601      → filter RX to BCM responses
0322F121D1D1D1D1 → ReadDataByIdentifier 0xF121
```

Response (ignition ON, raw ATCAF0+ATH1 format):

```
601 06 62 F1 21 01 86 63 00
```

Parse: strip 601 CAN ID prefix, read PCI 06 (6 data bytes), confirm service 62 + DID F1 21, extract data 01 86 63 = 99,939 km.

8.3 Performance Characteristics

All three reads are single-frame exchanges — one request, one response, no flow control required:

ECU	DID	CAN RTT	ELM327 AT setup	Total (BLE)
BCM	0xF121	~7 ms	~250 ms	~257 ms
DDE	0x46B1	~31 ms	~300 ms	~331 ms

ECU	DID	CAN RTT	ELM327 AT setup	Total (BLE)
HU	0x060A	~55 ms (CF delay n/a — SF)	~400 ms	~455 ms

Total for all three reads: approximately **1.0–1.2 seconds** via BLE ELM327. This is acceptable for a one-time read per connection.

8.4 Locale-Aware Display

The odometer value should be displayed in both metric (km) and imperial (mi) units, using the device locale for number formatting:

```
// km: locale-aware thousands separator (. in DE, , in UK/US/AU)
Text("\(km.formatted()) km")

// miles: km × 0.621371, rounded to nearest integer
let mi = Int((Double(km) * 0.621371).rounded())
Text("\(mi.formatted()) mi")
```

Swift’s `Int.formatted()` uses `Locale.current` automatically — no explicit locale configuration required. German users see 99.939 km, UK/US/AU users see 99,939 km.

9. Broader Implications

9.1 The Power of Passive Black-Box Analysis

This discovery illustrates a general principle that is highly effective for automotive protocol research: **let the OEM tool do the work, then read what it said.**

Bosch Grade-X represents decades of BMW engineering knowledge, embedded in a commercial product that has gone through OEM certification, extensive testing, and real-world deployment across millions of vehicles. When Grade-X queries a DID, it does so because that DID is meaningful, reliable, and supported across the relevant vehicle population.

By observing Grade-X’s queries — without running them ourselves, without guessing, without brute force — we gain immediate access to a curated, validated set of DIDs. The confirmation is built into the method: if Grade-X reads it and the value matches the physical instrument, the DID is confirmed by the highest possible authority (the OEM tool itself).

9.2 Generalisability to Other BMW-Architecture Vehicles

The DID addresses discovered here are almost certainly consistent across the BMW vehicle platform that the INEOS Grenadier shares:

- **0xF121** is the BMW standard odometer DID used in BCM across multiple model families (3-Series, 5-Series, X5 of the same generation)
- **0x46B1** is the DDE/DME engine management odometer DID, consistent across BMW diesel models using the B47/B57 engine family
- **0x060A** is the HU (MGU/HU-H3) infotainment odometer DID

Any researcher investigating a BMW 3-Series G20, X5 G05, or other vehicle sharing the same ECU generation is likely to find the same DIDs functional on their platform.

9.3 Limitations and Future Work

- **0x46B1 4-byte format:** The uint32 ceiling is 4,294,967,295 km — effectively unlimited. The uint24 ceiling used by BCM and HU (16,777,215 km) is similarly unlimited in any practical vehicle lifetime.
- **Petrol variant (DME):** DDE = diesel DDE. Petrol-engined Grenadiers use DME (Digital Motor Electronics). The DDE DID 0x46B1 may not be present on DME — confirmation requires a petrol vehicle with engine running and DME online. BCM 0xF121 and HU 0x060A are almost certainly universal.
- **MY26 variants:** Future model years may add FCM (Front Camera Module, 0x691/0x611) and DMS (Driver Monitoring System, 0x68E/0x60E). These ECUs may carry additional odometer DIDs worth investigating.
- **Real-time updates:** The odometer is read once per connection (ignition ON). For real-time mileage tracking during driving, a polling interval would be needed. The current implementation does not poll continuously — this is intentional (the vehicle is always operated with ignition in accessory position during diagnostic sessions).

10. Conclusion

Three independent odometer DIDs on the INEOS Grenadier have been discovered and validated through entirely passive, legal, ethical black-box observation of a Bosch Grade-X OEM diagnostic session:

ECU	DID	Format	Confirmed Value
BCM	0xF121	uint24 BE (3 bytes)	99,939 km OK
DDE	0x46B1	uint32 BE (4 bytes)	99,939 km OK
HU	0x060A	uint24 BE (3 bytes)	99,939 km OK

All three are accessible in the UDS default session without security access or extended session escalation. All three agree exactly with the physical instrument cluster reading, confirmed by the vehicle owner.

The methodology — passive parallel CAN capture during a legitimate OEM diagnostic session — is a powerful, legal, and ethical approach to automotive protocol research. It is explicitly covered by vehicle owner rights recognised in EU, UK, US, and Australian law. It does not involve circumventing access controls, decompiling software, or any activity that could harm the vehicle.

The findings are implemented in the open-source `grenadier-obd2-ios` application (MIT licence), freely available to all INEOS Grenadier owners worldwide. Any owner with a standard Bluetooth OBD2 ELM327 adapter can now read their own odometer — in both kilometres and miles, cross-validated across three independent ECUs, with full RemoteLogger telemetry for fleet research — without requiring dealer access or proprietary tooling.

Appendix A — Raw Capture Evidence

Primary capture file: `dumps/grade_x_sessions/grenadier_raw_20260619_154716.log`

Duration: 841 s · 243,075 frames · 2026-06-19

Key frames containing odometer value 0x018663 = 99,939 km:

```
(1781877084.204185)  can0  601    [8]  06 62 F1 21 01 86 63 00  ← BCM DID 0xF121
(1781877084.204185)  can0  612    [8]  07 62 46 B1 00 01 86 63  ← DDE DID 0x46B1
```

```
(1781877594.260165) can0 612 [8] 21 01 21 17 00 01 86 63 ← DDE freeze frame CF1
Secondary capture file: dumps/grade_x_sessions/grenadier_raw_20260619_152801.log
(1781873482.260165) can0 603 [8] 06 62 06 0A 01 86 63 AA ← HU DID 0x060A
```

Appendix B — UDS Request Frames Used in Application

Formatted as 8-byte padded CAN frames (ATCAF0, D1 fingerprint):

```
BCM odometer request: 681#0322F121D1D1D1D1 → expects: 601#0662F12101 8663 00
DDE odometer request: 692#032246B1D1D1D1D1 → expects: 612#076246B100 018663
HU odometer request: 683#032206 0AD1D1D1D1 → expects: 603#066206 0A01 8663 AA
```

Appendix C — Conversion Reference

km	hex (uint32 BE)	miles
99,939	00 01 86 63	62,107
100,000	00 01 86 A0	62,137
200,000	00 03 0D 40	124,274
500,000	00 07 A1 20	310,686
1,000,000	00 0F 42 40	621,371

Formula: `miles = km × 0.621371` (rounded to nearest integer for display)

End of document

This research is published under the grenadier-canlab project (MIT licence). All findings are shared freely for the benefit of the Grenadier owner community. No finding described herein enables harmful modifications to vehicle safety systems.